

OMEGA 3.0

A Portable, Memory-First Autonomous Cognitive Substrate

System Flowchart and Detailed Description of Operation

Patent Paperwork Supplement

Inventor: Noah Lee

DeRidder, Louisiana

Status: Patent Pending

System version reflected herein: Omega 3.0 (May 2026)

Abstract

Disclosed is a portable, memory-first autonomous cognitive substrate (100) in which cognition is anchored in durable structured memory rather than in the transient context window of a large language model. The substrate is organized as five vertically-separated layers — OMEGA (110), DEEP SLEEP (210), GLOBAL (310), META (410), and AUTONOMOUS (510) — operating in parallel with a user/system-initiated BRAIN LOOP (610). Each layer has explicit permissions and explicit prohibitions. Capability flows upward only through validation; governance flows downward only through policy. No layer is permitted to promote itself.

The substrate advances on a deterministic phase-clocked cadence driven by two parallel clocks: an EPOCH CLOCK (700) that rotates the layers in a fixed order, and a CADENCE CLOCK (710) that fires pressure-driven sub-epoch dialog pursuits. The action layer fires only when the planning layer has explicitly commanded it, via a rationale-bound action packet (930) that names the goal of record, the tool chosen, and the success definition. Every outcome is bundled and recorded (950), and the reflection layer ingests those outcomes alongside open gap-signal entries (970) to drive evidence-traceable goal formation. Capability emergence — i.e., the authoring of new tools — is reserved to the governance layer and is gated three ways: by health verdict (1110), by per-cycle budget (1120), and by registry-existence dedup (1130). A single inference call per qualifying cycle authors a candidate tool specification, which is then independently evaluated by a Guardian decision authority (1300) before registration. All structural change is auditable and reversible.

Field of the Invention

This invention pertains to architectures for autonomous cognitive systems. More specifically, it pertains to a memory-first substrate for bounded autonomy, in which inference is treated as replaceable tissue and intelligence is treated as a structural property of the system rather than as a function of model scale.

Background

Conventional autonomous-agent systems built atop large language models tend to conflate inference with intelligence, attempt to learn online inside the same loop they serve from, and bolt safety constraints on top of agents that have no internal notion of restraint. Three failure modes recur in such systems: (i) open-loop action — the system acts without recording why it acted or whether the action satisfied a goal; (ii) synthetic goal fabrication — when reflection produces no signal, the system invents goals from nothing in order to keep moving; and (iii) reflexive fallback — when the planning layer cannot find a tool for a goal, it silently substitutes a generic search, papering over the capability gap rather than recording it.

These failure modes are not artifacts of insufficient model scale. They are architectural. A system without a durable rationale-bound contract between planning and action cannot tell whether an outcome corresponded to a goal. A system without an evidence-anchored intake into goal formation will invent goals when no signal supports any. A system without a gated, deduped capability-emergence pipeline will either grow new tools uncontrollably or, in the absence of growth, will paper over gaps. The system disclosed herein addresses each of these failure modes architecturally, not by instruction.

Brief Description of the Drawings

Four figures accompany this disclosure:

- FIG. 1 depicts the five-layer cognitive architecture of the substrate, with the explicit may/may-not boundaries that govern each layer and the asymmetric directional flow of capability and governance.
- FIG. 2 depicts the epoch-cycle rotation by which the layers fire on the heartbeat clock, including the two gating conditions under which the action layer is permitted to fire and the same-epoch inline fire that closes the loop within a single cycle.
- FIG. 3 depicts the closed-loop feedback path: the rationale-bound action packet authored by the planning layer, the outcome bundle written by the action layer, and the closed-loop intake by which the reflection layer turns recorded outcomes into traceable goals.
- FIG. 4 depicts the tool-synthesis pipeline by which new capabilities enter the substrate: deterministic detection of capability gaps, three-gate dispatch, Guardian-bound spec decision, and reversible registration.

FIG. 1 — Five-Layer Cognitive Architecture

FIG. 1 — FIVE-LAYER COGNITIVE ARCHITECTURE

Capability flows upward through validation; governance flows downward through policy. No layer self-promotes.

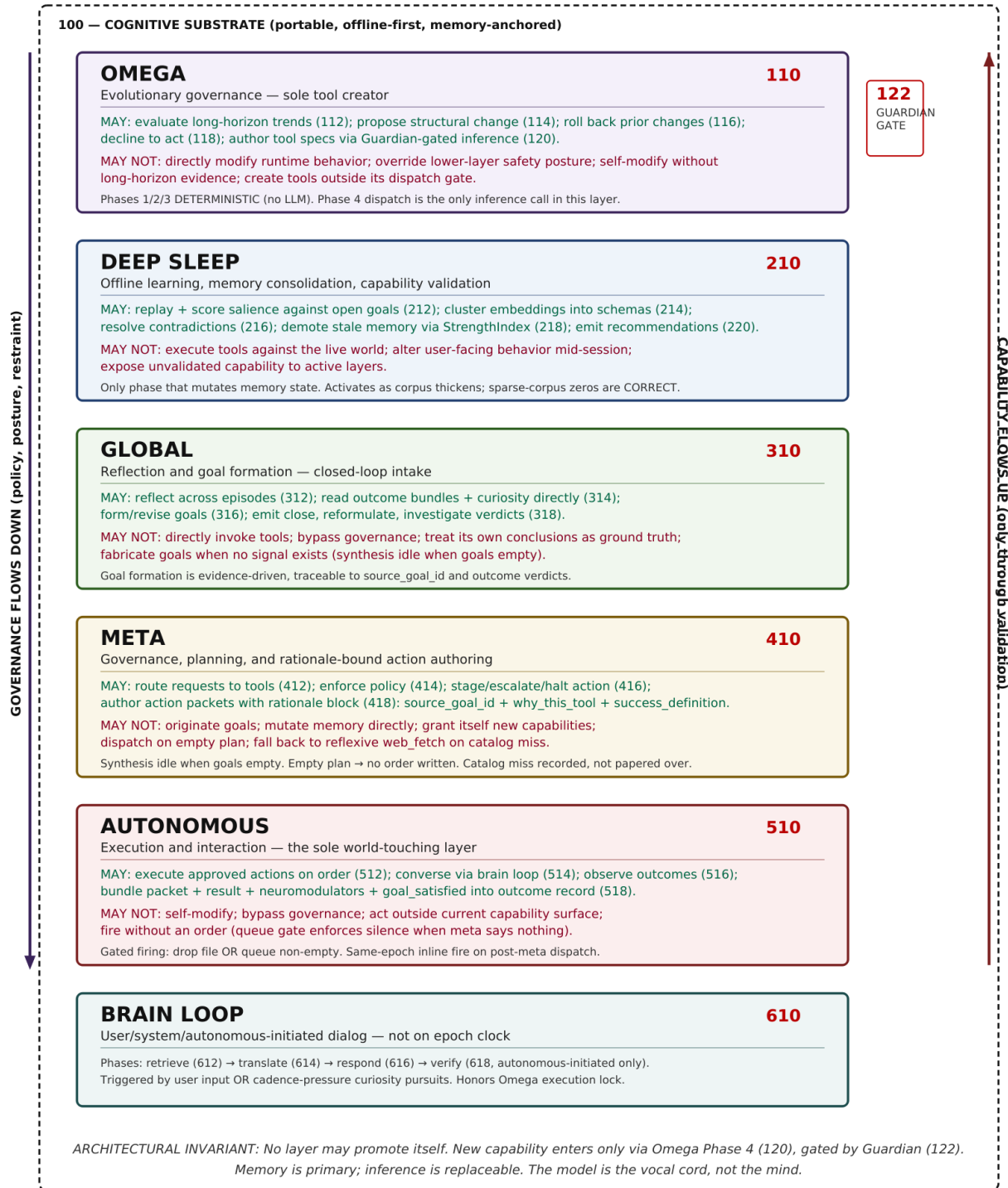


FIG. 1 — *Five-Layer Cognitive Architecture (vertical permission/prohibition stack).*

Reference Designator Legend for FIG. 1

Ref.	Designation
100	Cognitive substrate as a whole — portable, offline-first, memory-anchored.
110	OMEGA layer — evolutionary governance; sole authority for tool creation.
112	Long-horizon trend evaluation function of OMEGA.
114	Structural-change proposal authoring function.
116	Rollback function: revoke any prior committed proposal.
118	Decline-to-act capacity (the system may choose to do nothing).
120	Tool specification authoring via Guardian-gated inference (the only LLM call in this layer).
122	Guardian gate over OMEGA inference output.
210	DEEP SLEEP layer — offline learning and consolidation.
212	Salience-scoring replay against open goals.
214	Schema integration (embedding clustering, cosine threshold ≥ 0.75 , ≥ 3 members).
216	Contradiction reconciliation, routed through the Guardian seam.
218	StrengthIndex-based active forgetting (demotion, not erasure).
220	Recommendation emission to journal for OMEGA intake.
310	GLOBAL layer — reflection and goal formation.
312	Cross-episode reflection function.
314	Closed-loop intake reading outcome bundles and open curiosity entries directly.
316	Goal formation and revision function.
318	Verdict emission: close, reformulate, investigate, learn, fix, boost, rotate, deepen, strengthen.
410	META layer — governance, planning, action-packet authoring.
412	Tool routing function (selection from the catalog).
414	Policy enforcement function.
416	Action staging, escalation, or halting.
418	Rationale-block authoring (source_goal_id + why_this_tool + success_definition).
510	AUTONOMOUS layer — the sole world-touching execution surface.
512	Order execution function (drains the META drop file).
514	Conversation via the BRAIN LOOP.
516	Outcome observation.
518	Outcome bundle authoring: packet + result + neuromodulators + goal_satisfied.

Ref.	Designation
610	BRAIN LOOP — user/system/autonomous-initiated dialog; not on the epoch clock.
612	Perception, classification, and memory recall phase.
614	Translation phase: structured-response shaping from posture and retrieved references.
616	Response delivery phase.
618	Verification phase (fires only when the brain loop was autonomous-initiated).

Detailed Description of FIG. 1

Referring to FIG. 1, the cognitive substrate **(100)** is organized as six functional regions: five vertically-separated layers and one parallel dialog channel. Each layer has both a set of operations it MAY perform and a set of operations it MAY NOT perform. The architecture is, therefore, a safety argument before it is a technical one: restraint is encoded in the geometry of the system, not in instructions issued at runtime.

The OMEGA layer **(110)** is the system's evolutionary governance authority. It may evaluate long-horizon trends **(112)**, propose structural change **(114)**, roll back prior changes **(116)**, decline to act **(118)**, and author new tool specifications by way of a Guardian-gated inference call **(120)**. It may not modify runtime behavior directly, override the safety posture of any lower layer, self-modify without long-horizon evidence, or create tools outside of its single permitted dispatch gate. OMEGA Phases 1, 2, and 3 are wholly deterministic — they execute no inference whatsoever. Inference enters this layer only at Phase 4, and only when all three dispatch gates **(122)** have passed.

The DEEP SLEEP layer **(210)** is the substrate's offline-learning region. It consolidates memory by replay-driven salience scoring **(212)**, integrates new schemas from embedding clusters **(214)**, reconciles contradictions through the Guardian seam **(216)**, and demotes stale memory entries via the StrengthIndex **(218)**. It emits journal entries and recommendations for OMEGA intake **(220)**. It may not execute any tool against the live world, alter user-facing behavior mid-session, or expose unvalidated capability to active layers. Note that on a fresh, sparse memory corpus, this layer correctly produces zero schemas, zero contradictions resolved, and zero forgotten entries — its gates are biologically calibrated and activate only as the corpus thickens.

The GLOBAL layer **(310)** is the substrate's reflection and goal-formation authority. It reflects across episodes **(312)**, reads outcome bundles and the open-curiosity store directly **(314)**, forms and revises goals **(316)**, and emits verdicts **(318)**. It is forbidden to invoke tools directly, bypass governance, or treat its own conclusions as ground truth. Critically, when no observation evidence supports any goal, GLOBAL emits no goals; the system does not fabricate domain rotation or invent goals from nothing.

The META layer **(410)** is the substrate's planning authority. It routes requests to tools **(412)**, enforces policy **(414)**, stages and may halt action **(416)**, and authors each action packet with a rationale block **(418)**. It may not originate goals (those come from GLOBAL), mutate memory directly, grant itself new capabilities, dispatch on an empty plan, or fall back to reflexive web search when the catalog lacks a tool

for a goal. When the catalog cannot satisfy a goal, META records the capability miss; the failure is preserved as a signal for OMEGA, not papered over.

The AUTONOMOUS layer (510) is the sole world-touching execution surface. It executes approved actions (512), engages in conversation through the BRAIN LOOP (514), observes outcomes (516), and bundles each cycle's packet + result + neuromodulator updates + goal-satisfied verdict into an outcome record (518). It may not self-modify, bypass governance, or act outside its current capability surface. It may not fire absent an explicit order from META; when META has nothing to dispatch, AUTONOMOUS remains silent and the world is not touched.

The BRAIN LOOP (610) operates in parallel to the epoch-clocked layers and is not itself driven by the epoch clock. It is invoked by user input, by cadence-pressure curiosity pursuits, or by autonomous-initiated reflection. Its four phases — retrieve (612), translate (614), respond (616), and (when autonomous-initiated) verify (618) — share the same memory substrate, identity anchors, and verification machinery as the autonomous loop. The BRAIN LOOP honors the OMEGA execution lock and therefore cannot fire while OMEGA is dispatching a structural change.

The directional asymmetry shown in FIG. 1 is foundational. Governance flows downward only — higher layers may constrain lower ones, but never bypass them. Capability flows upward only through validation — a new function cannot be granted to a lower layer except through the OMEGA Phase 4 dispatch, gated by the Guardian (122). No layer is permitted to promote itself.

FIG. 2 — Epoch-Clocked Cognitive Rotation

FIG. 2 — EPOCH CYCLE: PHASE-CLOCKED COGNITIVE ROTATION

One ~7-minute epoch. Thinking layers fire every cycle. The action layer fires only when commanded.

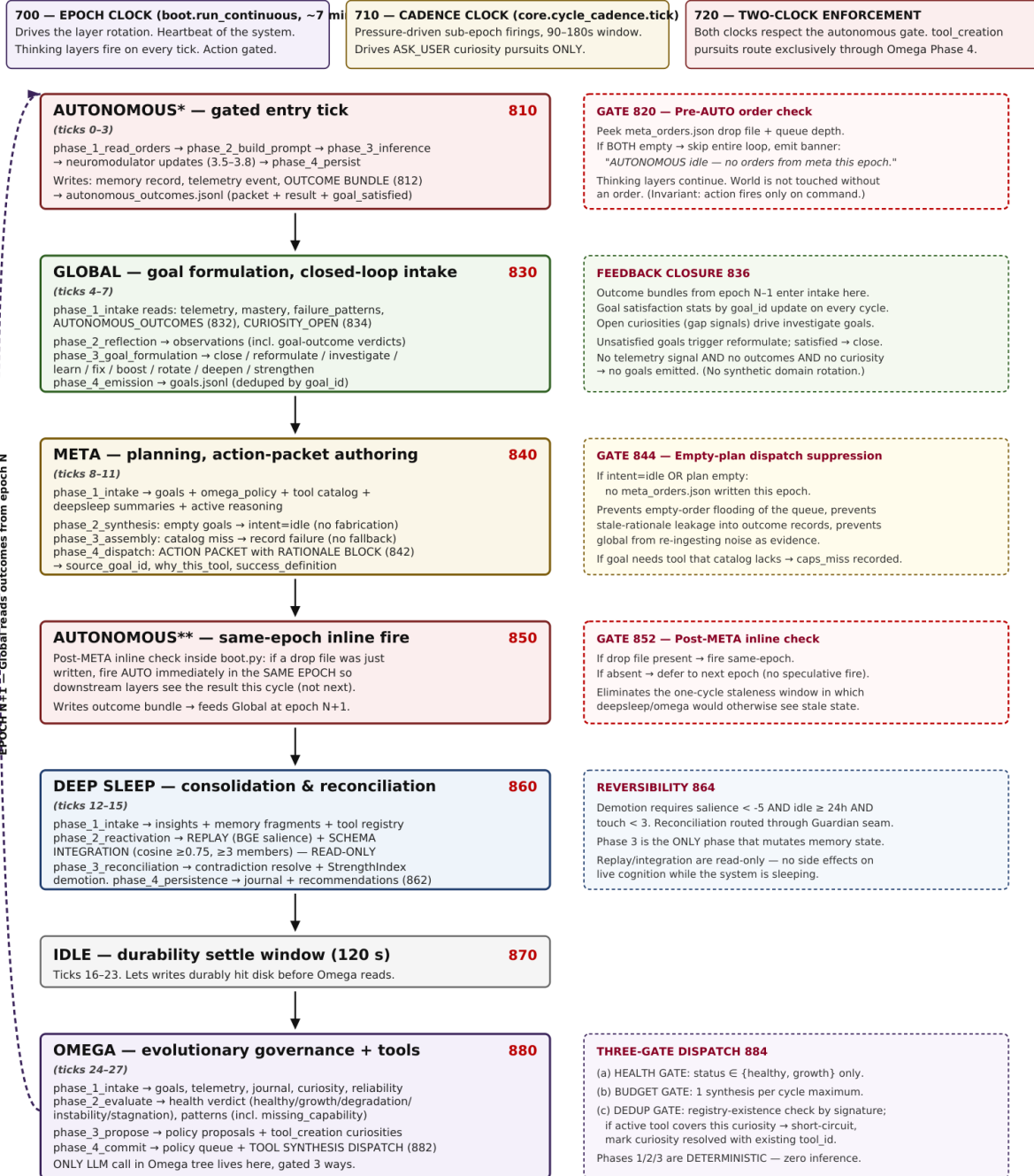


FIG. 2 — Phase-clocked epoch rotation (~7 minutes per cycle). The action layer fires only on command.

Reference Designator Legend for FIG. 2

Ref.	Designation
700	Epoch clock — drives the layer rotation, approximately one cycle per seven minutes.
710	Cadence clock — pressure-driven sub-epoch firings of ASK_USER curiosity pursuits, window 90–180 s.
720	Two-clock enforcement: both clocks respect the autonomous gate; tool-creation pursuits route exclusively through OMEGA Phase 4.
810	AUTONOMOUS* — the gated entry tick, ticks 0–3 of each epoch.
812	Outcome bundle written to autonomous_outcomes.jsonl.
820	Pre-AUTO order gate: skip the entire autonomous tick unless meta_orders.json drop file or queue is non-empty.
830	GLOBAL — goal formulation and closed-loop intake, ticks 4–7.
832	Closed-loop intake of autonomous outcomes.
834	Closed-loop intake of open curiosity entries.
836	Feedback closure: outcome bundles from epoch N–1 enter intake; no signal yields no goal.
840	META — planning and action-packet authoring, ticks 8–11.
842	Action packet rationale block: source_goal_id, source_goal_kind, source_goal_summary, source_goal_priority, why_this_tool, success_definition, capability_evidence.
844	Empty-plan dispatch suppression: when intent=idle or plan empty, no order is written.
850	AUTONOMOUS** — same-epoch inline fire if META dispatched.
852	Post-META inline check: fire same-epoch on drop file presence; defer otherwise.
860	DEEP SLEEP — consolidation and reconciliation, ticks 12–15.
862	Recommendation emission to deep_sleep_recommendations.jsonl.
864	Reversibility constraints on demotion: salience < -5 AND idle ≥ 24 h AND touch < 3.
870	IDLE — 120-second durability settle window, ticks 16–23.
880	OMEGA — evolutionary governance and tool creation, ticks 24–27.
882	Tool-synthesis dispatch (the sole inference call in the OMEGA tree).
884	Three-gate dispatch: health, budget, dedup.

Detailed Description of FIG. 2

Referring to FIG. 2, the substrate advances on two parallel clocks. The EPOCH CLOCK (**700**) is the heartbeat. It rotates the layers through a fixed sequence — AUTONOMOUS* (gated) → GLOBAL → META → AUTONOMOUS** (post-META inline) → DEEP SLEEP → IDLE → OMEGA — approximately every seven minutes. The CADENCE CLOCK (**710**) fires sub-epoch pursuits driven by composite cognitive pressure; its window tightens to 90 seconds under high pressure and relaxes to 180 seconds when the system is quiet. The cadence clock drives ASK_USER dialog pursuits exclusively. Tool-creation pursuits

flow only through OMEGA Phase 4 (720) — there is no second pathway by which new tools may enter the system.

The thinking layers — GLOBAL, META, DEEP SLEEP, and OMEGA — fire on every epoch tick. The action layer (AUTONOMOUS) fires only when META has commanded it. This separation between thinking and acting is enforced by the pre-AUTO order gate (820). At the beginning of each epoch, the gate peeks at both the meta_orders.json drop file and the order queue. If both are empty, the entire autonomous loop is skipped and a banner is emitted: "AUTONOMOUS idle — no orders from meta this epoch." The thinking layers proceed regardless. The world is not touched without an order.

When the autonomous layer does fire (810), it executes four primary phases (read orders → build prompt → inference → persist), interleaved with four neuromodulator updates (dopamine, plasticity, strength, stress). The final persistence phase writes three records: a memory record, a telemetry event, and the outcome bundle (812). The outcome bundle is the substrate's key feedback artifact, described in detail in connection with FIG. 3.

The GLOBAL layer (830) then aggregates evidence from prior cycles. Two of its intake sources are new in version 3.0: autonomous outcomes (832) and open curiosities (834). These together constitute the feedback closure (836). Outcome bundles from epoch N–1 update goal-satisfaction statistics by goal_id; unsatisfied goals trigger reformulation; satisfied ones trigger closure. Open curiosity entries — gap signals from failure thresholds, deep-sleep recommendations, and OMEGA-detected missing capabilities — drive investigate goals. When no signal supports any goal, GLOBAL emits no goal. This is the architectural fix for the synthetic-goal-fabrication failure mode.

The META layer (840) next consumes goals.jsonl, the omega_policy, the tool catalog (with deprecated and retired tools filtered), and the deep-sleep summaries. Its phase-2 synthesis is gated: when goals are empty, intent is set to idle and no domain is invented. Its phase-3 assembly attempts to bind tool calls against the catalog; on a catalog miss, the failure is recorded as a capability-miss signal rather than being papered over by a reflexive web search. Its phase-4 dispatch (842) writes the meta_orders.json drop file with a rationale block containing source_goal_id, source_goal_kind, source_goal_summary, source_goal_priority, why_this_tool, success_definition, and capability_evidence. When the plan is empty, no order is written (844) — preventing empty-order flooding of the queue and stale-rationale leakage into outcome records.

A critical architectural feature is the post-META inline check (852). Immediately after META completes, the substrate re-tests the drop-file condition; if META has just dispatched, AUTONOMOUS fires again (850) within the same epoch. This eliminates the one-cycle staleness window in which DEEP SLEEP and OMEGA would otherwise read pre-execution state. Outcomes generated in the same-epoch inline fire become available to GLOBAL in epoch N+1.

The DEEP SLEEP layer (860) performs consolidation, schema integration, contradiction reconciliation, and StrengthIndex-based demotion. Replay and integration are read-only; only phase 3 of this layer

mutates memory state, and demotion is subject to compound reversibility constraints (864). Its phase 4 writes the journal entry, deep-sleep recommendations (862), and truncates replay history.

The IDLE window (870) is a 120-second durability settle, after which the OMEGA layer (880) reads the journal. OMEGA evaluates system health, proposes config-change and capability-emergence candidates, and — when and only when the three dispatch gates (884) all pass — fires the sole inference call permitted in the OMEGA tree (882). The dispatch process is detailed in connection with FIG. 4.

FIG. 3 — Closed-Loop Cognitive Feedback

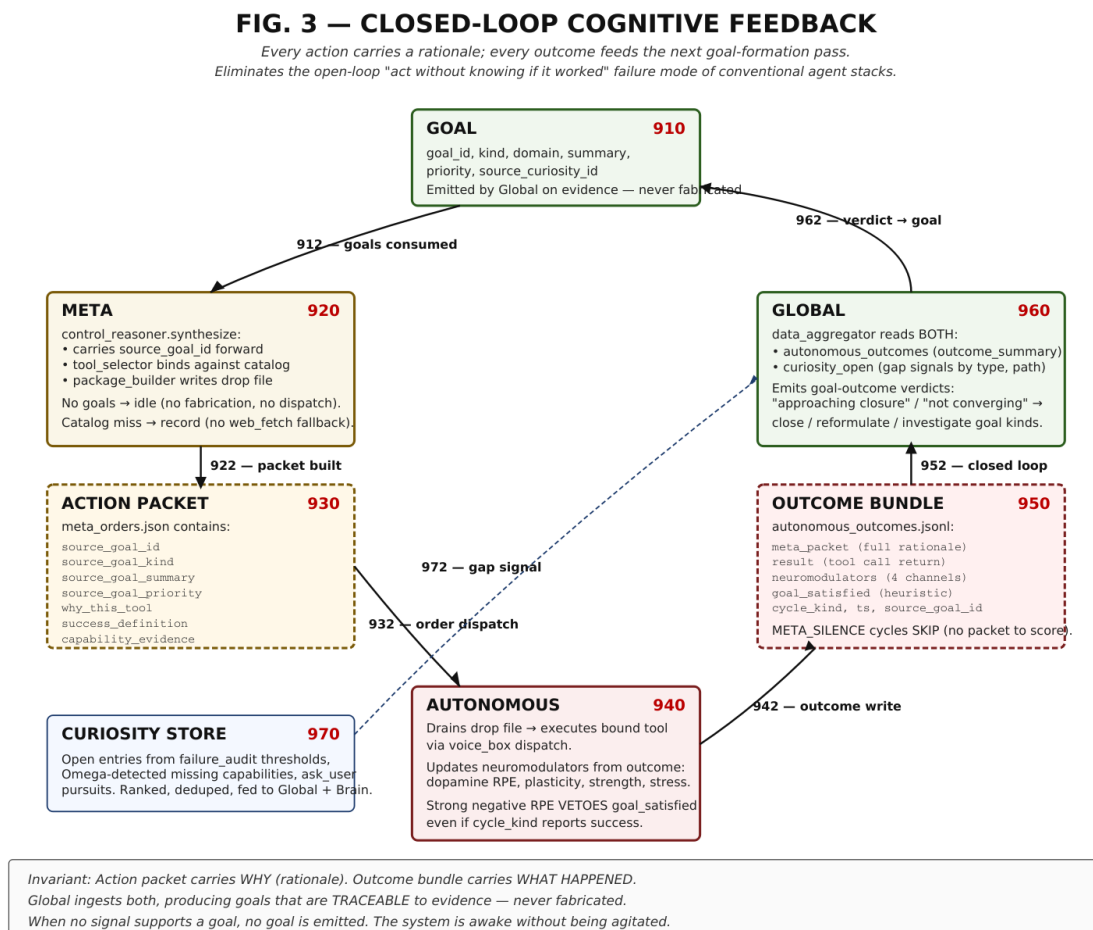


FIG. 3 — Rationale-bound action and outcome-bound goal formation — the closed feedback loop.

Reference Designator Legend for FIG. 3

Ref.	Designation
910	Goal record stored in goals.jsonl, deduped by goal_id.
912	Goal consumption arc — META reads goals.jsonl at epoch intake.
920	META packet authoring: control_reasoner.synthesize + tool_selector.select_and_bind + package_builder.build_and_dispatch.
922	Packet construction arc — once goals and a matching tool exist, the packet is built.
930	Action packet drop file (meta_orders.json) containing the seven-field rationale block.
932	Dispatch arc — meta_orders.json drop triggers AUTONOMOUS execution.
940	AUTONOMOUS execution via voice_box dispatch; neuromodulator updates from outcome.
942	Outcome write arc — bundle written to autonomous_outcomes.jsonl.

Ref.	Designation
950	Outcome bundle: full packet + result + neuromodulators + goal_satisfied verdict + cycle metadata.
952	Closed-loop intake arc — GLOBAL reads outcome bundles directly at intake.
960	GLOBAL closed-loop intake (data_aggregator).
962	Verdict arc — GLOBAL emits next-cycle goals (close, reformulate, investigate).
970	Curiosity store — gap signals from failure_audit thresholds, OMEGA-detected missing capabilities, and ask_user pursuits.
972	Gap-signal arc — open curiosities feed GLOBAL intake directly.

Detailed Description of FIG. 3

FIG. 3 depicts the closed-loop feedback path that distinguishes this substrate from conventional open-loop agent stacks. The cycle begins at the GOAL record (910), which carries a goal_id, kind, domain, summary, priority, and optionally a source_curiosity_id pointing back to the gap signal that motivated the goal. Goals are emitted by GLOBAL on observation evidence — never on confabulation.

The META layer (920) consumes goals (912) and produces the action packet (930). The action packet is a durable artifact written to meta_orders.json. Its rationale block contains seven fields: source_goal_id, source_goal_kind, source_goal_summary, source_goal_priority, why_this_tool, success_definition, and capability_evidence. The packet is, in effect, the planning layer's testimony — a record of which goal this action is meant to satisfy, why this tool was selected, and what would count as success.

The AUTONOMOUS layer (940) executes the bound tool via voice_box dispatch (932). Four neuromodulator channels update from the outcome — dopamine reward-prediction error, plasticity, strength, and stress. A strong negative dopamine RPE vetoes the goal_satisfied verdict even if the surface cycle_kind reports success; this prevents bad cycles from being counted as wins.

The execution produces the outcome bundle (950), written to autonomous_outcomes.jsonl (942). The outcome bundle is the substrate's key feedback artifact. It contains the full meta_packet (with the rationale block intact), the result of the tool call, the four neuromodulator channels, the goal_satisfied heuristic verdict, the cycle_kind, the timestamp, and a back-reference to source_goal_id. META_SILENCE cycles — those in which META wrote no order — skip the outcome bundle entirely, because there is no packet to evaluate.

The GLOBAL layer (960) closes the loop by reading both the outcome bundles (952) and the open curiosity store (972) at intake. Goal-outcome verdicts (962) — "approaching closure" or "not converging" — flow into the next cycle's goal formation as close, reformulate, or investigate kinds.

The CURIOSITY STORE (970) is a parallel gap-signal channel. Entries arrive from failure_audit when a pattern accumulates past threshold; from OMEGA when Phase 3 detects a missing-capability pattern; and from the brain loop when a user-facing pursuit was registered. Each entry is scored using a

canonical scoring contract and may be ranked, deduped, or short-circuited if registry already provides a covering tool.

The result is that every action is traceable to a goal and every goal is traceable to evidence. When no signal supports a goal, no goal is emitted. The system is, in the inventor's phrasing, "awake without being agitated."

FIG. 4 — Tool-Synthesis Pipeline (Sole Capability-Emergence Path)

FIG. 4 — TOOL SYNTHESIS PIPELINE (OMEGA, SOLE TOOL CREATOR)

Capability emergence is gated, deduped, Guardian-approved, and reversible. Inference fires once per qualifying cycle, never speculatively.

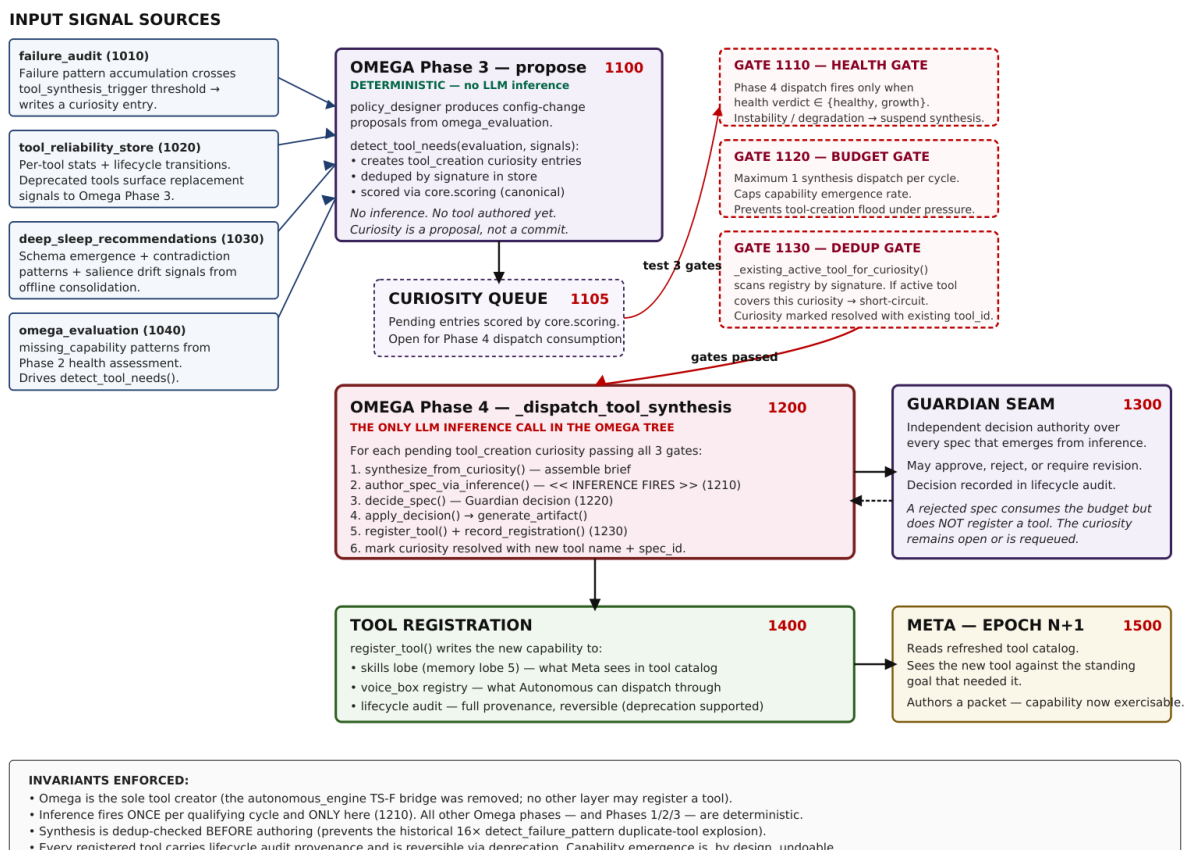


FIG. 4 — Three-gate dispatch with Guardian-bound spec decision and reversible registration.

Reference Designator Legend for FIG. 4

Ref.	Designation
1010	failure_audit: failure-pattern accumulation that, on threshold crossing, writes a curiosity entry.
1020	tool_reliability_store: per-tool statistics and lifecycle transitions that surface deprecated-tool replacement signals.
1030	deep_sleep_recommendations.jsonl: schema emergence, contradiction patterns, and salience-drift signals from offline consolidation.
1040	omega_evaluation: missing-capability patterns produced by OMEGA Phase 2 health assessment.
1100	OMEGA Phase 3: deterministic proposal authoring. Produces config-change proposals and tool-creation curiosity entries; no inference.
1105	Curiosity queue: pending tool_creation entries scored via canonical contract.

Ref.	Designation
1110	Health gate: dispatch fires only when status $\in$ {healthy, growth}.
1120	Budget gate: maximum 1 synthesis dispatch per cycle.
1130	Dedup gate: registry-existence check by signature; short-circuit when an active tool already covers the curiosity.
1200	OMEGA Phase 4: tool-synthesis dispatch (the sole permitted inference call in the OMEGA tree).
1210	author_spec_via_inference — the single inference call.
1220	Guardian decision over the authored spec.
1230	register_tool + record_registration: writes to skills lobe, voice_box registry, and lifecycle audit.
1300	Guardian seam: independent decision authority over every synthesized spec.
1400	Tool registration record: skills lobe + voice_box registry + lifecycle audit.
1500	META at epoch N+1: reads the refreshed catalog and sees the new capability.

Detailed Description of FIG. 4

FIG. 4 depicts the substrate's tool-synthesis pipeline — the sole pathway through which new capabilities enter the system. The pipeline begins with four input signal sources: failure_audit (**1010**), tool_reliability_store (**1020**), deep_sleep_recommendations (**1030**), and omega_evaluation (**1040**). Each of these surfaces a different kind of capability-gap evidence: accumulated failures crossing a threshold, deprecated tools requiring replacement, schema or contradiction patterns produced offline, and missing-capability patterns detected in OMEGA's health assessment.

OMEGA Phase 3 (**1100**) processes these signals deterministically — no inference is performed at this stage. The policy_designer produces config-change proposals from the omega_evaluation, and detect_tool_needs() converts opportunity signals into tool-creation curiosity entries (**1105**). Each entry is deduped by signature in the curiosity store and scored via the canonical scoring contract. Phase 3 is, in effect, a proposal layer: a curiosity is a candidate, not a commit.

OMEGA Phase 4 dispatches synthesis only when three gates simultaneously permit. The health gate (**1110**) requires the system's status to be healthy or in a growth phase; degradation, instability, or stagnation suspend synthesis. The budget gate (**1120**) permits a maximum of one synthesis dispatch per cycle, capping capability-emergence rate even under pressure. The dedup gate (**1130**) calls _existing_active_tool_for_curiosity(), which scans the registry by signature; if an active tool already covers the curiosity, the dispatch short-circuits and the curiosity is marked resolved with the existing tool_id.

Only when all three gates pass does the pipeline enter the dispatch proper (**1200**). The dispatch is a six-step pipeline: synthesize_from_curiosity assembles a brief; author_spec_via_inference (**1210**) performs the single inference call permitted in the OMEGA tree; decide_spec (**1220**) routes the candidate spec

through the Guardian seam (1300); apply_decision proceeds to generate_artifact only when the Guardian approves; register_tool + record_registration (1230) commits the new capability; and the originating curiosity is marked resolved with the new tool's name and spec_id.

The Guardian seam (1300) is an independent decision authority over every spec that emerges from inference. It may approve, reject, or require revision. A rejected spec consumes that cycle's synthesis budget but does NOT register a tool; the originating curiosity remains open or is requeued. The Guardian decision is recorded in the lifecycle audit.

Tool registration (1400) writes the new capability to three locations: the skills lobe (memory lobe 5), which is what META sees in its tool catalog at the next epoch; the voice_box registry, which is what AUTONOMOUS can actually dispatch through; and the lifecycle audit, which carries full provenance and supports deprecation. Capability emergence is, by design, reversible.

At epoch N+1, META (1500) reads the refreshed catalog and sees the new tool against the standing goal that motivated its creation. The capability has thereby entered the substrate not by edict but through an evidence-gated, deduped, Guardian-approved, reversible path.

Operational Sequence — One Full Epoch Trace

The following sequence illustrates one representative epoch of substrate operation, traced through the figures. It demonstrates how the architectural elements described above interlock to produce closed-loop, evidence-traceable cognition.

Epoch N

Step 1. The autonomous gate (820) is tested. If both the drop file and the queue are empty, AUTONOMOUS* (810) is skipped this cycle; the thinking layers continue.

Step 2. GLOBAL (830) performs intake. Outcome bundles (832) from epoch N–1 update goal-satisfaction statistics; open curiosities (834) surface gap signals. Reflection produces observations and verdicts. Where evidence justifies, goals are emitted to goals.jsonl (910); where no signal supports a goal, no goal is written.

Step 3. META (840) reads goals.jsonl and the tool catalog. If no goals exist, intent is set to idle and no order is written (844). If a goal exists and a matching tool exists, package_builder writes meta_orders.json (930) with the seven-field rationale block (842). If the goal exists but no matching tool can be found, a catalog_miss is recorded and no order is written; the gap will resurface to OMEGA next cycle.

Step 4. The post-META inline check (852) tests the drop file. If META just dispatched, AUTONOMOUS** (850) fires in the same epoch. Tool execution updates neuromodulators; the outcome bundle (950) is written, including the full rationale block and the goal_satisfied verdict.

Step 5. DEEP SLEEP (860) performs consolidation (read-only replay and schema integration), reconciliation (the only mutating phase), and persistence (journal + recommendations (862)). Demotion is subject to reversibility constraints (864).

Step 6. IDLE (870) settles for 120 seconds, allowing the prior writes to durably reach disk before OMEGA reads them.

Step 7. OMEGA (880) performs deterministic phases 1, 2, and 3 — state collection, health evaluation, and proposal authoring (including new tool_creation curiosity entries (1105)). Phase 4 tests the three dispatch gates (884). If all pass, one synthesis dispatches: author_spec_via_inference (1210) fires the sole permitted LLM call in the OMEGA tree; the Guardian seam (1300) decides; if approved, register_tool (1230) commits.

Epoch N+1

Step 8. GLOBAL (830) ingests autonomous_outcomes from epoch N — outcome_summary updates goal-satisfaction statistics by goal_id. Reformulate or close goal mappings fire if warranted.

Step 9. META (840) sees the new tool registered in epoch N now present in the skills lobe catalog, against the standing goal that needed it. It writes meta_orders.json binding the new tool.

Step 10. AUTONOMOUS (810) fires, executes the new tool, writes its outcome bundle. The loop completes: a capability gap was sensed, became a curiosity, was scored, gated, Guardian-approved, registered, and exercised — all while the substrate produced a complete provenance audit trail.

Invariants Enforced (Claims-Ready Summary)

The following invariants are enforced architecturally — not by instruction, not by prompt, not by a runtime check that may be skipped or overridden — and are therefore proper subjects of claims to the structural novelty of the substrate.

Action-on-order invariant.

The autonomous execution layer (510) fires if and only if the planning layer (410) has authored an action packet (930) that has been received via the pre-AUTO order gate (820) or the post-META inline gate (852). Absent an order, the autonomous layer remains silent and the world is not touched. Thinking layers continue independently.

Same-epoch execution invariant.

When the planning layer dispatches mid-epoch, the action layer fires within the same epoch (850, 852), so that downstream consolidation (860) and governance (880) read post-execution state in cycle N rather than cycle N+1.

Synthesis-idle invariant.

When the reflection layer (310) emits no goals, the planning layer (410) sets intent=idle and writes no action packet (844). The substrate does not fabricate goals or synthetic domain rotation in the absence of supporting evidence.

Empty-plan suppression invariant.

When the planning layer cannot bind a tool to a goal, no action packet is written (844). The capability miss is recorded as a signal for governance, not papered over by a reflexive search.

Rationale-bound action invariant.

Every action packet (930) carries a rationale block containing source_goal_id, source_goal_kind, source_goal_summary, source_goal_priority, why_this_tool, success_definition, and capability_evidence (842). No action may be authored without these fields.

Outcome-bundling invariant.

Every executed cycle writes an outcome bundle (950) to autonomous_outcomes.jsonl bundling the full action packet, the tool result, the four neuromodulator channels, the goal_satisfied verdict, the cycle_kind, the timestamp, and the source_goal_id back-reference. META_SILENCE cycles, having no packet, are skipped.

Closed-loop intake invariant.

The reflection layer (310) reads outcome bundles (832) and the open-curiosity store (834) directly at intake. Goal formation is therefore grounded in recorded evidence, not in blind generation.

Tool-creation singularity invariant.

The governance layer (110) is the sole authority for tool creation. No other layer registers tools. The legacy autonomous_engine TS-F bridge has been removed; there exists no alternate registration pathway.

Three-gate dispatch invariant.

Tool-synthesis dispatch (1200) fires only when the health gate (1110), the budget gate (1120), and the dedup gate (1130) all pass. The maximum dispatch rate is one synthesis per epoch.

Guardian-bound capability invariant.

Every spec emerging from inference is independently evaluated by the Guardian seam (1300) before registration. A rejected spec consumes the budget but does NOT register a tool.

Reversibility invariant.

Every registered tool carries lifecycle-audit provenance and is reversible by deprecation. Every memory demotion is subject to compound reversibility constraints (864). No structural change is undoable only at runtime.

Deterministic-governance invariant.

Phases 1, 2, and 3 of the governance layer are deterministic — no inference runs. Inference enters the governance tree at one location only: Phase 4 dispatch (1210). All other governance reasoning is reproducible from input state.

Canonical-scoring invariant.

All score computations across producers use the canonical scoring contract (core.scoring). Hand-rolled magic numbers are forbidden.

Negative-RPE veto invariant.

A strong negative dopamine reward-prediction error vetoes the goal_satisfied verdict, even when cycle_kind reports success. Bad cycles are not counted as wins.

Memory-primacy invariant.

When inference output conflicts with verified structured memory, memory wins. The model is replaceable; the memory substrate is not.

Notes for Patent Counsel

The figures herein use color and dashed/solid distinctions to aid readability during prosecution and review. Formal filing may require conversion to black-and-white line art per the rules of the relevant patent office; the underlying SVG source files are preserved alongside this document and convert cleanly. The reference numerals used (100s for Cognitive Substrate / OMEGA region; 200s DEEP SLEEP; 300s GLOBAL; 400s META; 500s AUTONOMOUS; 600s BRAIN LOOP; 700s clocks; 800s epoch-cycle steps; 900s closed-loop feedback; 1000s tool-synthesis inputs; 1100s OMEGA Phase 3 + gates; 1200s OMEGA Phase 4; 1300s Guardian; 1400s registration; 1500s next-epoch META) follow a consistent block-allocation scheme to permit later expansion (e.g., 130, 132, 134 sub-features within OMEGA) without renumbering.

The terminology used throughout this document — "rationale-bound action packet," "outcome bundle," "closed-loop intake," "three-gate dispatch," "Guardian seam," "memory primacy" — is consistent with the inventor's formal white paper and the in-system source-code naming. Counsel may wish to confirm terminology before claim drafting.

This document is a supplement intended to accompany the formal Omega 3.0 white paper (filed concurrently). It is not itself a patent specification; it is structured to provide clear, numbered, figure-anchored support for the specification's detailed-description section, and to map cleanly onto independent and dependent claim language.